Physical Constants as Eisenstein Fourier Coefficients: Observational Identities and a Wheeler–DeWitt Derivation of $\Omega_\Lambda = 2\pi/9$

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Abstract

We report several arithmetic identities linking observed dimensionless physical constants to Fourier coefficients of normalized Eisenstein series. Specifically, the leading term of the fine structure constant $\alpha^{-1} = 2/|B_2| + 4/|B_4| + \ldots = 132 + \ldots$ equals the q^1-coefficient of $(E_4 - E_2)/2$, the Wright Brothers "Bernoulli ladder" entries $L_k = 2k/|B_{2k}| \in \{12, 120, 252, 240, 132\}$ are exactly half the absolute values of normalized Eisenstein series q^1-coefficients, and the dark energy density parameter admits a functional-equation dual form $\Omega_\Lambda = 16|\zeta(-1)|\zeta(2)/\pi = 2\pi/9 \approx 0.6981$. This latter value is obtained independently from a quantum cosmology argument: in minisuperspace Wheeler–DeWitt theory, the cosmic wave function $\Psi(a)$ satisfies temporal Dirichlet–Neumann boundary conditions (Dirichlet at Big Bang, Neumann at de Sitter asymptotic), whose 1D zero-point energy zeta-regularizes to $\zeta_{\neg 2}(-1) = +1/12$. Combined with the Cohen–Kaplan–Nelson holographic bound, this yields the observed Ω_Λ at the 2% level, consistent with Planck 2018 within 1.8σ. We discuss the connection of these identities to modular forms in physics (heterotic strings, moonshine, topological strings) and to the Kreimer–Connes Hopf algebra formulation of perturbative QED. The observations suggest that certain dimensionless physical constants may be expressible as evaluations of modular-form constructions involving ζ special values at integer arguments — a specific physical instance of the Deligne–Beilinson period conjectures. We explicitly list limitations, including the absence of rigorous first-principles derivations for most identities, and identify concrete test paths via future cosmological surveys (Euclid, DESI, LSST) and precision particle measurements (Belle II, Muon $g-2$).

1 Introduction

Two long-standing problems in theoretical physics are the origin of dimensionless fundamental constants, such as the fine structure constant $\alpha^{-1} = 137.035999\ldots$ [1], and the numerical value of the cosmological constant, $\rho_\Lambda \approx 5.24 \times 10^{-10}$ J/m^3, corresponding to $\Omega_\Lambda = 0.6847 \pm 0.0073$ (Planck 2018) [2]. The first lacks a predictive derivation from any current theory; the second is notoriously 10^{123} orders of magnitude smaller than naive QFT estimates [3].

This work reports observational identities connecting these constants (and others, including the anomalous muon magnetic moment $\Delta a_\mu \approx 251 \times 10^{-11}$ [4]) to Fourier coefficients of normalized Eisenstein series $E_{2k}(\tau)$ and related modular-form objects. While modular forms have long appeared in mathematical physics — notably heterotic string theory [13], Monstrous Moonshine [14], Mathieu Moonshine [15], topological string invariants, and black hole state counting — their systematic use to *determine numerical values of low-energy dimensionless constants* has been essentially absent from the literature. The present work proposes such a program.

The second main result is a quantum cosmology argument for Ω_Λ. We show that if the minisuperspace Wheeler–DeWitt wave function $\Psi(a)$ satisfies Dirichlet boundary conditions at the Big Bang ($a = 0$) and Neumann conditions at future infinity ($a \to \infty$), then the associated 1D

zero-point energy regularizes via $\zeta_{\neg 2}(-1) = +1/12$. Combined with the Cohen–Kaplan–Nelson holographic bound [6], this yields

$$\Omega_\Lambda = \frac{2\pi}{9} \approx 0.6981 \tag{1}$$

as a parameter-free prediction, consistent with Planck 2018 at the 1.8σ level.

Our approach differs from the canonical "string landscape" treatment of the cosmological constant problem [12] in that it produces a specific numerical value rather than invoking anthropic selection over a multiverse.

1.1 Structure of the paper

Section 2 fixes notation. Section 3 states the main identities. Section 4 presents the Wheeler–DeWitt temporal DN derivation of $\Omega_\Lambda = 2\pi/9$. Section 5 places the results in the context of existing modular-form physics. Section 6 lists concrete testable predictions. Section 7 honestly assesses limitations.

2 Notation and preliminaries

We use the standard normalized Eisenstein series of weight $2k \geq 4$:

$$E_{2k}(\tau) = 1 - \frac{4k}{B_{2k}} \sum_{n=1}^{\infty} \sigma_{2k-1}(n)\, q^n, \quad q = e^{2\pi i \tau} \tag{2}$$

where B_{2k} are the Bernoulli numbers and $\sigma_s(n) = \sum_{d|n} d^s$. The Fourier coefficient of q^1 in E_{2k} is therefore $-4k/B_{2k}$.

The quasi-modular Eisenstein series $E_2(\tau)$ is defined by the same formula with $k = 1$ (so the q^1 coefficient is -24). It fails modular invariance by

$$E_2\left(\tfrac{a\tau+b}{c\tau+d}\right) = (c\tau + d)^2 E_2(\tau) - \frac{6ic(c\tau + d)}{\pi}, \tag{3}$$

and its modular completion is $E_2^*(\tau) = E_2(\tau) - 3/(\pi \operatorname{Im} \tau)$.

The Riemann zeta function satisfies

$$\zeta(1-2k) = -\frac{B_{2k}}{2k}, \qquad \zeta(2k) = \frac{(-1)^{k+1}(2\pi)^{2k} B_{2k}}{2(2k)!}, \tag{4}$$

so that $|\zeta(-1)| = 1/12$, $|\zeta(-3)| = 1/120$, $|\zeta(-5)| = 1/252$, etc.

We define the *Wright Brothers Bernoulli ladder* by

$$L_k \equiv \frac{2k}{|B_{2k}|} = \frac{1}{|\zeta(1-2k)|}. \tag{5}$$

Its first entries are

$$(L_1, L_2, L_3, L_4, L_5) = (12,\ 120,\ 252,\ 240,\ 132). \tag{6}$$

At $k = 6$, $L_6 = 32760/691$ is non-integer due to the irregular prime 691 dividing the numerator of B_{12} [26].

3 Main observations

3.1 Ladder-Eisenstein identity

Identity 1 (Ladder-Eisenstein). *For $k \geq 1$,*

$$L_k = \frac{2k}{|B_{2k}|} = \frac{1}{2}\big|\textit{coefficient of } q^1 \textit{ in } E_{2k}\big|. \tag{7}$$

Proof. The q^1-coefficient of E_{2k} is $-4k/B_{2k}$. Taking absolute value and dividing by 2 yields $2k/|B_{2k}| = L_k$. □

Although mathematically elementary, this identity gives a systematic modular-form source for the Bernoulli ladder entries that appear throughout the Wright Brothers framework.

3.2 α^{-1} leading term from $(E_4 - E_2)/2$

Theorem 1 (α^{-1} modular identity)**.**

$$\frac{E_4(\tau) - E_2(\tau)}{2} = 1 + 132\,q + O(q^2) \tag{8}$$

where $132 = 2/|B_2| + 4/|B_4|$ *equals the leading arithmetic part of the Wright Brothers formula for* α^{-1}*.*

Proof. $E_4 = 1 + 240q + O(q^2)$ and $E_2 = 1 - 24q + O(q^2)$, so $(E_4 - E_2)/2 = 1 + (240 - (-24))/2 \cdot q + O(q^2) = 1 + 132q + O(q^2)$. The equality $132 = 12 + 120 = 2/|B_2| + 4/|B_4|$ is immediate from $|B_2| = 1/6$ and $|B_4| = 1/30$. □

3.3 Muon $g - 2$ anomaly: triple representation

The experimentally observed muon anomalous magnetic moment difference $\Delta a_\mu = (251 \pm 59) \times 10^{-11}$ [4, 5] admits three independent "modular" representations of the numerical value 252:

Observation 1 (Triple representation of 252)**.**

1. *(Bernoulli ladder):* $L_3 = 6/|B_6| = 252$
2. *(Eisenstein difference):* $(E_8 - E_2)/2 = 1 + 252\,q + O(q^2)$
3. *(Ramanujan discriminant):* $\tau(3) = 252$*, where* $\Delta(\tau) = \sum \tau(n)q^n$ *is the cusp form of weight 12*

Proof. (1) $L_3 = 6/|1/42| = 252$. (2) $E_8 = 1 + 480q + O(q^2)$ (since $E_8 = E_4^2$ at weight 8), so $(E_8 - E_2)/2 = 1 + (480 + 24)/2 \cdot q + O(q^2) = 1 + 252q + O(q^2)$. (3) $\tau(3) = 252$ is a classical value of Ramanujan's tau function. □

The numerical coincidence $\Delta a_\mu \approx 252$ (in units of 10^{-11}) was previously noted [33]. The new observation is that this number simultaneously equals a single-Eisenstein ladder entry, an Eisenstein difference, and a cusp form coefficient — three independent modular-form objects.

3.4 Non-trivial ladder additive identities

Identity 2 (Ladder additive relations)**.** *Within the Wright Brothers ladder, the following non-trivial identities hold:*

$$L_1 + L_2 = L_5 \qquad \textit{(i.e., } 12 + 120 = 132\textit{)} \tag{9}$$

$$L_1 + L_4 = L_3 \qquad \textit{(i.e., } 12 + 240 = 252\textit{)} \tag{10}$$

$$L_2 + L_5 = L_3 \qquad \textit{(i.e., } 120 + 132 = 252\textit{)} \tag{11}$$

These express non-trivial Bernoulli relations such as $|B_2|^{-1} + 2|B_4|^{-1} = 5|B_{10}|^{-1}$ (equivalently $6 + 60 = 66$), which is elementary once stated but has no obvious derivation from standard Bernoulli recursion formulas. The closure of the ladder under certain sums suggests hidden structure worth further investigation.

3.5 Ω_Λ as functional-equation dual

Identity 3 (Functional-equation form of Ω_Λ).

$$\Omega_\Lambda = \frac{2\pi}{9} = 8\pi B_2^2 = -\frac{16\,\zeta(-1)\,\zeta(2)}{\pi} = \frac{16\,|\zeta(-1)|\,\zeta(2)}{\pi}. \tag{12}$$

Proof. $B_2 = 1/6$, so $8\pi B_2^2 = 8\pi/36 = 2\pi/9$. For the zeta form: $\zeta(-1) = -1/12$, $\zeta(2) = \pi^2/6$, so $-16\zeta(-1)\zeta(2)/\pi = 16 \cdot (1/12)(^2/6)/\pi = 16\pi/72 = 2\pi/9$. □

This form is notable in that it combines ζ at a negative integer ($\zeta(-1)$, algebraic via Bernoulli) and a positive integer ($\zeta(2)$, a period involving π^2), with an explicit $1/\pi$ regularization factor. Such a combination uses both sides of the Riemann functional equation.

3.6 E_2 quasi-modular connection

The $1/\pi$ factor in Identity 3 is intriguingly parallel to the modular completion of E_2:

$$E_2^*(\tau) = E_2(\tau) - \frac{3}{\pi\,\mathrm{Im}\,\tau}, \tag{13}$$

in which $3/(\pi y)$ is the quasi-modular correction. We cannot yet derive Ω_Λ directly from evaluation of E_2^* at a specific point, but the shared $1/\pi$ structure suggests a connection. A known special value is $E_2(i) = 3/\pi$ [25], from which $E_2^*(i) = 0$. We note $\Omega_\Lambda = 2\pi/9 \neq (E_2(i))^{-1}$, so no trivial identification exists.

4 Wheeler–DeWitt derivation of Ω_Λ

We now present an independent derivation of $\Omega_\Lambda = 2\pi/9$ from quantum cosmology.

4.1 Minisuperspace Wheeler–DeWitt equation

The Wheeler–DeWitt equation [7, 8] in minisuperspace, for a flat FRW universe with inflaton ϕ, reduces to a 1D Schrödinger-like equation for the wave function $\Psi(a, \phi)$:

$$\left[-\frac{\hbar^2}{2}\frac{\partial^2}{\partial a^2} + V(a) + \hat{H}_\phi\right]\Psi(a,\phi) = 0. \tag{14}$$

The potential $V(a)$ incorporates curvature, cosmological constant, and matter contributions [9].

4.2 Temporal Dirichlet–Neumann boundary conditions

We propose that $\Psi(a)$ satisfies the following boundary conditions:

Proposition 1 (Temporal DN boundary).

$$\Psi(a=0) = 0 \quad \textit{(Dirichlet at Big Bang)}, \tag{15}$$

$$\partial_a \Psi|_{a\to\infty} \textit{ free} \quad \textit{(Neumann at de Sitter asymptotic)}. \tag{16}$$

Justification: (i) $a = 0$ corresponds to the Big Bang singularity, at which classical geometry breaks down. The requirement $\Psi(0) = 0$ is consistent with the no-boundary proposal of Hartle and Hawking [10] interpreted in Lorentzian signature, and with the tunneling proposal of Vilenkin [11]. (ii) In the far future, the universe asymptotes to de Sitter with constant Λ, and $\Psi(a) \sim a^{-3/2} e^{\pm i a^3\sqrt{\Lambda}/(3\hbar)}$ in WKB approximation, consistent with a Neumann-like (free-oscillation) boundary condition.

4.3 1D DN zero-point energy

A 1D harmonic cavity with length L and Dirichlet–Neumann boundary conditions has modes $k_n = (2n-1)\pi/(2L)$, $n = 1, 2, \ldots$. The zero-point energy, regularized via zeta function, is

$$E_0^{\mathrm{DN}} = \frac{\hbar c\pi}{4L} \sum_{n=1}^{\infty} (2n-1)_{\mathrm{reg}} = \frac{\hbar c\pi}{4L} \cdot \zeta_{\neg 2}(-1) \tag{17}$$

where

$$\zeta_{\neg 2}(s) \equiv \sum_{\text{odd } n \geq 1} n^{-s} = (1 - 2^{-s})\zeta(s). \tag{18}$$

At $s = -1$:

$$\zeta_{\neg 2}(-1) = (1-2)\zeta(-1) = (-1)(-1/12) = +\frac{1}{12}. \tag{19}$$

The positive sign is essential. In contrast, an NN or DD cavity uses $\zeta(-1) = -1/12$, giving attractive (negative) Casimir energy. The DN case yields positive Casimir energy, appropriate for dark energy ($\rho_\Lambda > 0$).

4.4 Cohen–Kaplan–Nelson bound

The CKN bound [6] states that vacuum energy in a region of size L satisfies

$$\rho_{\mathrm{vac}}(L) \lesssim M_P^2/L^2 = \rho_P \big(\ell_P/L\big)^2. \tag{20}$$

Applied to the observable universe with $L = \ell_H$ (Hubble length), and assuming saturation with coefficient $|\zeta_{\neg 2}(-1)| = 1/12$ from the DN structure, we obtain

$$\rho_\Lambda = \frac{1}{12} \cdot \rho_P \cdot \big(\ell_P/\ell_H\big)^2. \tag{21}$$

4.5 Deriving $\Omega_\Lambda = 2\pi/9$

Using $\rho_P = \hbar c/\ell_P^4$ and $\ell_P^2 = \hbar G/c^3$:

$$\rho_\Lambda = \frac{1}{12} \cdot \frac{c^4}{G\ell_H^2}. \tag{22}$$

The observational definition gives

$$\rho_\Lambda = \Omega_\Lambda \cdot \rho_c = \Omega_\Lambda \cdot \frac{3c^4}{8\pi G\ell_H^2}, \tag{23}$$

where $\ell_H = c/H_0$. Equating the two expressions:

$$\frac{1}{12} = \frac{3\Omega_\Lambda}{8\pi} \quad \Longrightarrow \quad \boxed{\Omega_\Lambda = \frac{2\pi}{9}}. \tag{24}$$

Numerically, $2\pi/9 = 0.69813$, agreeing with Planck 2018 observation 0.6847 ± 0.0073 at the 1.8σ level (2.0%). Notably, the prediction is independent of H_0, since ℓ_H cancels in the dimensionless ratio. This means the tension between Planck and SH0ES values of H_0 does not affect the Ω_Λ prediction.

5 Context: modular form physics

Modular forms have long been important in theoretical physics. Among the most notable applications:

- **Heterotic string theory**: Partition functions of $E_8 \times E_8$ and $SO(32)$ heterotic strings involve Eisenstein series and theta functions of even unimodular lattices [13]. In particular, the theta function of the E_8 lattice equals E_4: $\theta_{E_8}(\tau) = E_4(\tau)$ [25].
- **Monstrous Moonshine**: The j-function Fourier coefficients are dimensions of Monster group representations, proved by Borcherds [14] via vertex operator algebras.
- **Mathieu Moonshine**: The K3 elliptic genus involves a mock modular form whose Fourier coefficients decompose into dimensions of M_{24} representations [15, 16].
- **Topological strings**: Partition functions on Calabi–Yau threefolds generate quasi-modular forms encoding Gopakumar–Vafa invariants [17, 18].
- **Black hole microstates**: $\mathcal{N} = 4$ dyon degeneracies are Fourier coefficients of Igusa cusp forms [19] and mock Jacobi forms [20].
- $\mathcal{N} = 4$ **SYM S-duality**: The partition function is $SL_2(\mathbb{Z})$-invariant [21].
- **Kreimer–Connes renormalization**: Perturbative QED Feynman integrals evaluate to multiple zeta values and polylogarithms [22, 23, 24].

The present work is complementary to these. Existing applications predominantly concern higher-dimensional string theory, topological sectors, or formal BPS-state counting. The identities in Section 3 instead connect low-energy *numerical values* of dimensionless constants — in the 4D physical world — to modular-form objects.

This is closest in spirit to the Kreimer–Connes framework, where perturbative QED computations produce arithmetic constants. However, Kreimer–Connes does not predict specific values like $\alpha^{-1} = 137.036$ or $\Omega_\Lambda = 0.698$; it provides the framework for computing them loop-by-loop, but the values themselves emerge from ordinary quantum field theory. The present work proposes a direct modular-form "dictionary" that bypasses the loop expansion and gives closed-form arithmetic expressions.

5.1 Relation to Deligne–Beilinson period conjectures

The Deligne conjecture [27] and Beilinson conjectures [28] predict that critical L-values of motives factor as products of periods and algebraic numbers. The form

$$\Omega_\Lambda = \frac{16|\zeta(-1)|\zeta(2)}{\pi} \tag{25}$$

is reminiscent of such a factorization: $\zeta(-1)$ is rational (algebraic), $\zeta(2) = \pi^2/6$ is a period, and $1/\pi$ is a dimensional factor. If the Wheeler–DeWitt derivation is correct, Ω_Λ would be a physical instance of Beilinson's conjecture for an as-yet-unidentified motive.

6 Predictions and tests

We list testable predictions of this framework.

6.1 Cosmological surveys

- $\Omega_\Lambda = 2\pi/9$: Future cosmological surveys (Euclid [29], DESI [30], LSST [31]) will measure Ω_Λ at the $\sim 0.5\%$ level. Decisive test of $2\pi/9 = 0.69813$ versus current best-fit 0.6847.
- **Equation of state $w = -1$ exactly**: The derivation assumes ρ_Λ is constant, predicting $w_{\mathrm{DE}} = -1$ with no time variation. Recent DESI 2024 hints of $w(z) \neq -1$ [32] would, if confirmed, falsify this prediction.

6.2 Particle physics

- **Muon $g-2$ precision**: The identification $\Delta a_\mu \leftrightarrow \tau(3) = 252$ suggests precise normalization. Current Fermilab+BNL combined value of $(251 \pm 59) \times 10^{-11}$ [4] is within uncertainty.
- **Tau $g-2$**: If the generation-$\tau(n)$ correspondence holds, Δa_τ should be proportional to $\tau(5) = 4830$. Currently untestable at 10^{-2} precision; Belle II and future colliders may reach 10^{-4}.

6.3 Laboratory condensed matter

- **Dark energy length scale**: The prediction $\rho_\Lambda \approx 5.3 \times 10^{-10}$ J/m^3 corresponds to characteristic length $\ell_\Lambda = (\hbar c/\rho_\Lambda)^{1/4} \approx 88$ μm. Laboratory tests of the Casimir force and related vacuum effects at this scale (quartz bulk acoustic wave resonator thickness series) may reveal phase transition signatures [34].
- **Dirichlet–Neumann boundary Casimir**: The framework predicts a change in Casimir physics at boundaries with mixed DN structure. Commercial solidly-mounted resonators (SMR) provide such boundaries and have not been analyzed in this context.

7 Limitations

We honestly enumerate the limitations of this work:

1. **Mathematical triviality of the identities**: Identities like Identity 1 and Theorem 1 follow immediately from the definitions of Bernoulli numbers and Eisenstein series. The novelty lies in the *connection to physics*, not in mathematical content per se.
2. **Post-hoc nature of most observations**: The connections $\alpha^{-1} \leftrightarrow (E_4 - E_2)/2$ and $\Delta a_\mu \leftrightarrow \tau(3)$ are observed after the fact rather than predicted ab initio. Look-elsewhere effect cannot be fully controlled.
3. **CKN coefficient**: The Wheeler–DeWitt derivation of Ω_Λ assumes saturation of the CKN bound with coefficient 1/12, which we justify via the DN mode regularization. A rigorous first-principles derivation of this coefficient is not yet available.
4. **Strict Dirichlet issue**: A strict Dirichlet boundary at $a = 0$ in the mode decomposition leads to unnormalizable Bogoliubov coefficients; one must introduce a smoothing parameter (see [35]). The precise form of this smoothing is not unique.
5. **Quadrupole anomaly**: The prediction of low-ℓ CMB suppression from the DN boundary quantitatively underestimates the observed $\ell = 2$ quadrupole suppression ($\approx 40\%$) by an order of magnitude. Simple smoothed Dirichlet produces at most $\sim 10\%$ suppression [35].
6. **Mass hierarchy and mixing**: The framework does not address fermion masses, Yukawa couplings, or CKM/PMNS mixing, which have more than 10^{-7} observational precision but no analog of the Bernoulli-ladder-type arithmetic expression.

7. **Rigorous derivation from a Lagrangian**: No known QFT or gravity Lagrangian derives the observed identities from first principles. The Kreimer–Connes framework provides tools that may enable such a derivation, but this remains future work.

8 Discussion

The observations reported here suggest that certain dimensionless physical constants — notably α^{-1}, Ω_Λ, and Δa_μ — are representable as modular-form evaluations or functional-equation combinations of Riemann ζ special values. The status of these representations ranges from elementary identities (the ladder-Eisenstein correspondence) to speculative conjectures (the generation-Ramanujan-τ hypothesis).

The Wheeler–DeWitt derivation of $\Omega_\Lambda = 2\pi/9$ provides the most concrete link between quantum cosmology and modular-form arithmetic in this program. It is falsifiable at the 0.5% level by forthcoming cosmological surveys.

If these connections prove structural rather than accidental, they would suggest a deeper role for arithmetic geometry, and specifically for objects of the Langlands program and the Beilinson conjectures, in the determination of physical constants. The "Wright Brothers alphabet" — Bernoulli numbers, ζ special values, Ramanujan τ, and related modular data — would then constitute a systematic language for dimensionless parameters.

If they prove accidental, the program will fail specific predictions (especially $\Omega_\Lambda = 2\pi/9$ at $< 0.5\%$ precision and tau $g-2$), and the observations will be filed under numerical coincidence. Either outcome is scientifically productive.

9 Conclusion

We have reported several arithmetic identities linking observed dimensionless physical constants to Fourier coefficients of normalized Eisenstein series and to Riemann ζ special values. The most concrete results are:

(i) The Wright Brothers Bernoulli ladder $L_k = 2k/|B_{2k}|$ is exactly half the absolute value of the Eisenstein q^1-coefficient of E_{2k}.

(ii) The leading arithmetic part of the Wright Brothers formula for α^{-1}, equal to 132, coincides with the q^1-coefficient of $(E_4 - E_2)/2$.

(iii) The muon anomalous magnetic moment $\Delta a_\mu \approx 252 \times 10^{-11}$ has a triple modular representation: as L_3, as $(E_8 - E_2)/2|_{q^1}$, and as the Ramanujan discriminant coefficient $\tau(3)$.

(iv) The dark energy density parameter $\Omega_\Lambda = 2\pi/9 \approx 0.6981$ admits the functional-equation dual form $\Omega_\Lambda = 16|\zeta(-1)|\zeta(2)/\pi$.

(v) The same $\Omega_\Lambda = 2\pi/9$ arises independently from a Wheeler–DeWitt minisuperspace argument with temporal Dirichlet–Neumann boundary conditions combined with the Cohen–Kaplan–Nelson holographic bound.

These results are testable at the 0.5% level by future cosmological surveys. If borne out, they would provide a specific physical realization of the Deligne–Beilinson period conjectures within the context of dark energy and quantum cosmology.

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